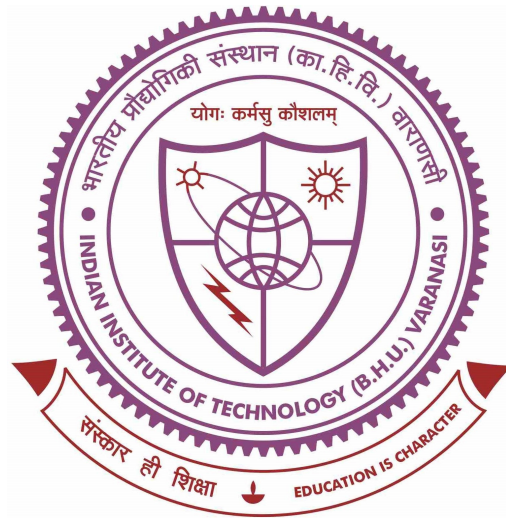


Nonlinear quantum phenomena in hybrid photon-magnon systems



Thesis submitted in partial fulfilment

for the Award of

INTEGRATED DUAL DEGREE (B.TECH.+ M.TECH.)

in

ENGINEERING PHYSICS

by

FIZAAN KHAN

DEPARTMENT OF PHYSICS

INDIAN INSTITUTE OF TECHNOLOGY

(BANARAS HINDU UNIVERSITY)

VARANASI – 221 005

ROLL NUMBER
21174011

YEAR OF SUBMISSION
2026

CERTIFICATE

It is certified that the work contained in the thesis titled **Nonlinear quantum phenomena in hybrid photon-magnon systems** by **Fizaan Khan** has been carried out under my/our supervision and that this work has not been submitted elsewhere for a degree.

It is further certified that the student has fulfilled all the requirements of Comprehensive Examination, Candidacy and SOTA for the award of **Integrated Dual Degree (B.Tech.+ M.Tech.)**.

Supervisors

Prof. Rajeev Singh

Prof. Biswanath Bhoi

Department of Physics
Indian Institute Of Technology
(Banaras Hindu University)
Varanasi – 221 005

DECLARATION BY THE CANDIDATE

I, **Fizaan Khan**, certify that the work embodied in this thesis is my own bona fide work and carried out by me under the supervision of **Prof. Rajeev Singh** and **Prof. Biswanath Bhoi**, from **January 2025** to **May 2026**, at the **Department of Physics**, Indian Institute of Technology (Banaras Hindu University), Varanasi. The matter embodied in this thesis has not been submitted for the award of any other degree/diploma. I declare that I have faithfully acknowledged and given credits to the research workers wherever their works have been cited in my work in this thesis. I further declare that I have not willfully copied any other's work, paragraphs, text, data, results, *etc.*, reported in journals, books, magazines, reports dissertations, theses, *etc.*, or available at websites and have not included them in this thesis and have not cited as my own work.

Date: April 22, 2026
Place: Varanasi, India



Signature of the Student

CERTIFICATE BY THE SUPERVISOR

It is certified that the above statement made by the student is correct to the best of my/our knowledge.

Supervisors

Prof. Rajeev Singh

Prof. Biswanath Bhoi

Department of Physics
Indian Institute Of Technology
(Banaras Hindu University)
Varanasi – 221 005

Signature of Head of Department

COPYRIGHT TRANSFER CERTIFICATE

Title of the Thesis: Nonlinear quantum phenomena in hybrid photon-magnon systems

Name of the Student: Fizaan Khan

COPYRIGHT TRANSFER

The undersigned hereby assigns to the Indian Institute of Technology (Banaras Hindu University), Varanasi all rights under copyright that may exist in and for the above thesis submitted for the award of the Integrated Dual Degree (B.Tech.+ M.Tech.) in Engineering Physics.

Date: April 22, 2026
Place: Varanasi, India

A handwritten signature in black ink, appearing to read 'Fizaan Khan', written over a light blue rectangular background.

Signature of the Student

Note: However, the author may reproduce or authorize others to reproduce material extracted verbatim from the thesis or derivative of the thesis for author's personal use provided that the source and the Institute's copyright notice are indicated.

ACKNOWLEDGEMENT

I'd like to thank my supervisor and mentor, Prof. Rajeev Singh, for his guidance and support throughout this project. His insights have been invaluable in shaping the direction of this research, while his faith in me in the toughest of times allowed this thesis to see its completion. I am also grateful to my co-supervisor, Prof. Biswanath Bhoi, for his expertise, constructive feedback and patient pedagogy that greatly enhanced the quality of this work. Finally, there's my parents, who made sure I could want for nothing, and stood by every phase and supported every new passion.

Contents

1	Background and Introduction	2
1.1	Background	2
1.2	Historical Development of Cavity Magnonics	3
1.2.1	The theoretical framework	3
1.2.2	Experimental Realisation and the Transition to Room Temperature	4
2	Mediated Coupling in Hybrid Photon-Magnon Systems	6
2.1	Motivation and Scope of the Study	6
3	Results and Discussion	9
4	Methods and Calculations	13
4.1	Design and Numerical Modeling of the Hybrid Photon-Magnon System .	13
4.2	Theoretical Formalism for Photon Mediated Magnon-Magnon Coupling .	15
4.3	Spatial distributions of microwave magnetic field distribution at the resonance peak of the hexagonal ring resonator (HRR)	18
5	Conclusion and Outlook	20
	Appendices	22
A	Derivation of transmission parameter S_{21}	23
B	Experimental Setup	26
B.1	Hardware	26

B.2	Software	28
B.2.1	Drivers	28
B.2.2	Stack	29
B.3	Data Analysis	29

List of Figures

1.1	Directions and goals for magnon hybrid systems. Taken from <i>Li et al.</i> (2020).[1]	2
1.2	Schematic of the proposed cavity magnonic system by Soykal and Flatté. Taken from <i>Soykal and Flatté</i> (2010).[2]	3
1.3	Schematic of the Huebl <i>et al.</i> (2013) (1.3a) and Zhang <i>et al.</i> (2014) (1.3b) experimental setups for demonstrating strong coupling in cavity magnonics. While Huebl <i>et al.</i> used a superconducting coplanar waveguide resonator at millikelvin temperatures, Zhang <i>et al.</i> achieved strong coupling at room temperature using a YIG sphere in a 3D copper cavity.	5
2.1	(a) The setup includes the two magnonic films placed on a pair of opposing sides of a copper hexagon (the photonic resonator). The ports are plugged into a vector network analyzer to both excite the microstrip line and also measure the transmission spectrum while a magnetic field is applied along the y -direction. (b) is the schematic of the model developed for this system. The spheres are bosonic oscillators with creation (annihilation) operator $\hat{b}_j^\dagger(\hat{b}_j)$. α_j denote intrinsic damping of the oscillators, and λ_j couple the oscillators to the traveling photon. The constants g_1 and g_2 denote the coupling of the magnonic cavities to the photonic cavity.	8
3.1	Observed (top row) and calculated (bottom row) S_{21} spectra of the system for corresponding selected thicknesses of YIG. The permalloy crossing is also seen to widen with a change in YIG thickness.	9

- 3.2 The blue crosses in (a) and (b) represent the coupling strengths of the YIG film to the resonator. The coupling strengths in (a) are for a system that does not include a permalloy film. Upon introduction of the permalloy in (b), the coupling strengths of the YIG have decreased by a fixed amount. The red dots in (b) mark g_1 (of permalloy), the coupling strengths of the permalloy to the resonator, and blue crosses are g_2 (of YIG), the coupling strengths of the YIG to the resonator. In (c), we plot the linear relationship between the two coupling strengths, $g_1(g_2)$ 10
- 4.1 A comparison of the magnonic coupling strengths of the (a) permalloy, (b) YIG, and (c) both to the resonator in an indirectly coupled configuration. The overlaid white curves are the real part of the eigenvalues of (4.6) at each value of H and were used to fit the model. 14
- 4.2 Simulated transmission spectra and field distributions of the hybrid magnon–photon–magnon system at 1100 Oe (~ 87 kA/m). (a) S-parameter spectrum showing the hybridized resonance (b) electric field distribution, and (c) magnetic (h -field) distribution at this resonance. The colors represent the strength of the electric and magnetic field in the respective field profile. 18
- 4.3 Spatial distributions of microwave magnetic field (h) distribution of HRR at resonance frequency $f = 5.29$ GHz. The colors represent the strength of the magnetic field. 19
- B.1 Top-down view of the sample, showing the hexagonal ring resonator (HRR) and the microstrip line (MSL) with SMA connectors. The YIG and permalloy films are mounted on opposite arms of the HRR, allowing for the study of photon-mediated magnon-magnon coupling. The magnetic field is applied perpendicular to the MSL, as denoted by the blue H vector. 27

B.2	Photograph of the experimental setup, showing the Rhode&Shwartz ZNLE18 vector network analyzer (left) connected to the sample via SMA connectors, and the Holmarc EM3000S electromagnet (right) used to apply the external magnetic field. The sample is placed in the gap of the magnet, allowing for precise control of the magnetic field while measuring the transmission spectra.	27
B.3	Overview of the codebase architecture, showing the main components and their interactions.	29

Chapter 1

Background and Introduction

1.1 Background

The emergence of cavity magnonics changed the investigation of magnetic resonance substantially — by coupling magnons (quantized spin waves) to electrodynamic fields, re-

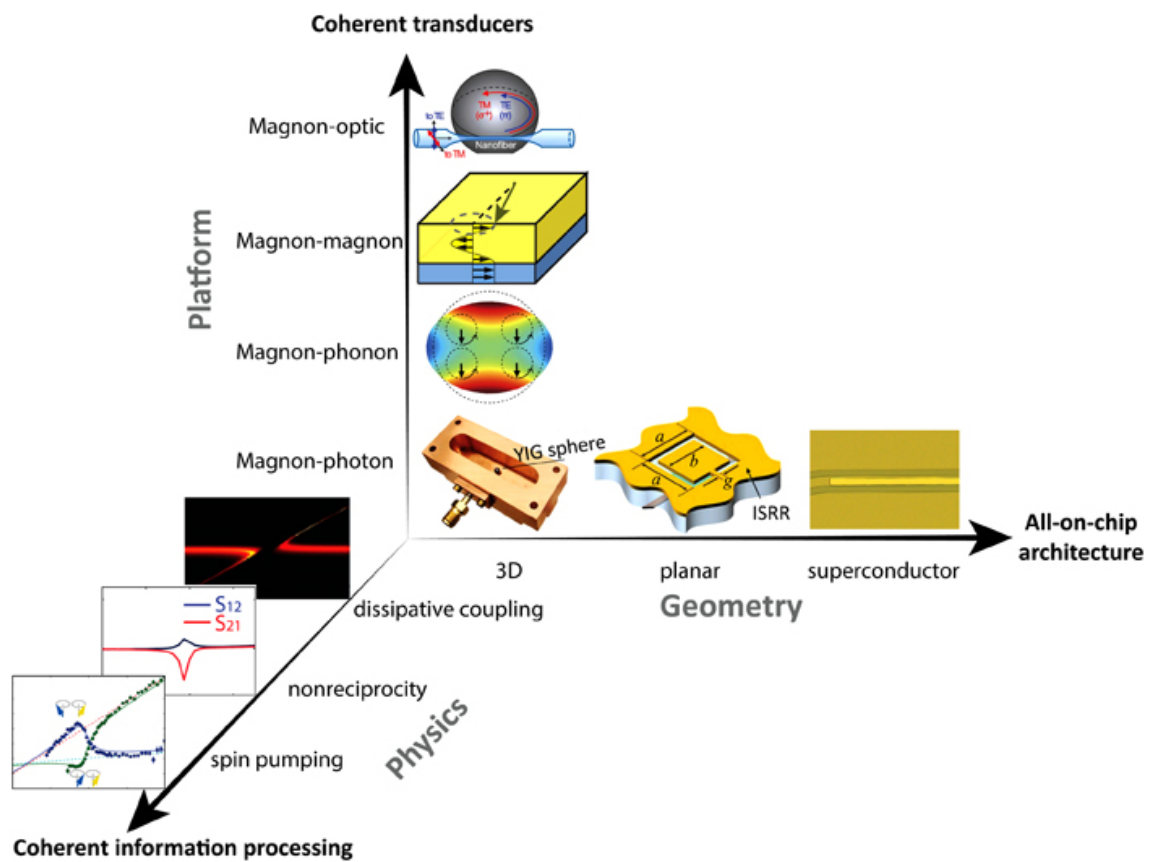


Figure 1.1: Directions and goals for magnon hybrid systems. Taken from *Li et al.* (2020).[\[1\]](#)

searchers found they could engineer hybridized states with a dual spin-photon character that neither subsystem could exhibit alone. This opened the door to new functionality: the coupled system behaves in ways that are simply inaccessible to its uncoupled components.

1.2 Historical Development of Cavity Magnonics

1.2.1 The theoretical framework

In 2010, Ö. O. Soykal and M. E. Flatté laid the theoretical groundwork for cavity magnonics by demonstrating that the macroscopic magnetic dipole moment of a ferromagnet could strongly interact with the electromagnetic field of a microwave cavity [2]. Prior to this, strong light-matter interactions were largely achieved via electric dipole interactions in atomic and solid-state systems. It is now known [3, 4] that by coupling a cavity photon to the collective uniform precession of spins in a ferromagnet—the Kittel mode—the effective coupling strength is en-

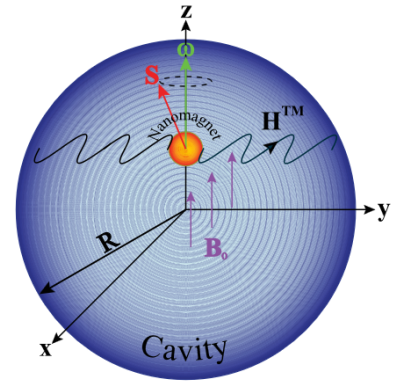


Figure 1.2: Schematic of the proposed cavity magnonic system by Soykal and Flatté. Taken from *Soykal and Flatté (2010)*. [2]

hanced by a factor of $\sqrt{N}$, where N is the total number of interacting spins. They proposed a nanomagnet-in-cavity system, as seen in Fig. 1.2, where a nanomagnet of an appropriate dimension to act as a macrospin is placed inside a microwave cavity, allowing for strong coupling between the cavity photons and the collective spin excitations (magnons) of the ferromagnet. Since solid-state ferromagnets possess exceptionally high spin densities, N can easily reach 10^9 [2]. This massive collective enhancement propels the system into the strong coupling regime, where the coherent energy exchange rate surpasses the dissipation rates of both the cavity and the magnetic subsystem. This seminal work highlighted magnetic materials as a robust, high-cooperativity alternative to traditional electric dipole systems for quantum electrodynamics (QED).

1.2.2 Experimental Realisation and the Transition to Room Temperature

The theoretical predictions of Soykal and Flatté were first experimentally validated in 2013 by Huebl *et al.*, who demonstrated strong coupling between a microwave resonator and a ferrimagnetic insulator [5]. In their pioneering setup (whose schematic is given in Fig. 1.3a), a gallium-doped yttrium iron garnet (YIG) crystal was coupled to a superconducting coplanar waveguide resonator. To suppress thermal noise and minimize dissipation, the experiment was conducted at millikelvin temperatures. By measuring the transmission spectrum, the team observed a clear avoided level crossing—the definitive signature of strong coupling—proving the viability of cavity magnonics in the quantum regime.

While Huebl *et al.* established the field, a major breakthrough in accessibility occurred the following year. In 2014, Zhang *et al.* successfully demonstrated the strong coupling of magnons and cavity microwave photons at room temperature [6]. By utilizing a highly polished, millimeter-scale YIG sphere placed inside a three-dimensional, high-conductivity copper cavity, they achieved strong coupling with exceptionally high cooperativity without the need for cryogenic cooling (Fig. 1.3b). This was made possible by the extremely low intrinsic damping of YIG and the large mode volume of the 3D cavity, which preserved the high Q -factor of the system.

This room-temperature realization significantly lowered the barrier to entry for experimental magnonics, accelerating the development of hybrid quantum systems and pushing them toward practical, scalable architectures. However, as these hybrid devices grow in complexity and begin to integrate multiple spatially separated magnetic elements, new interaction pathways inevitably emerge. While the direct coupling between a single magnon mode and a cavity photon is now well-established, the shared microwave environment inherently introduces indirect coupling mechanisms between otherwise isolated magnonic components. Understanding how the photonic cavity mediates these long-range interactions—and how such crosstalk can be leveraged or mitigated—forms the core natural motivation for the investigations detailed in the subsequent chapters of this thesis.

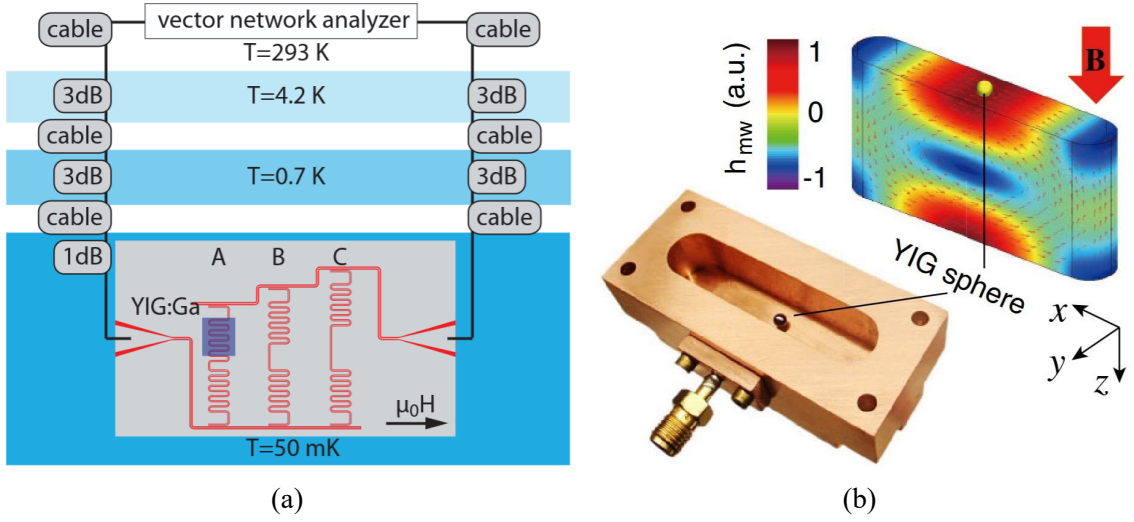


Figure 1.3: Schematic of the Huebl *et al.* (2013) (1.3a) and Zhang *et al.* (2014) (1.3b) experimental setups for demonstrating strong coupling in cavity magnonics. While Huebl *et al.* used a superconducting coplanar waveguide resonator at millikelvin temperatures, Zhang *et al.* achieved strong coupling at room temperature using a YIG sphere in a 3D copper cavity.

Fig. 1.1 illustrates the broad landscape of research directions and goals in the field of magnon hybrid systems. Here, we concern ourselves with magnon-photon coupling in planar geometries, which is a critical step toward scalable, on-chip quantum devices. The work done during the course of this thesis lead to the publication Khan *et al.* (2025) [7].

Chapter 2

Mediated Coupling in Hybrid Photon-Magnon Systems

2.1 Motivation and Scope of the Study

The advancement of hybrid quantum systems that integrate magnonic and photonic components has emerged as a critical frontier in the pursuit of scalable quantum information processing and next-generation microwave technologies[8, 9, 10, 11, 12]. In such systems, magnons—the quanta of spin waves—interact coherently with microwave photons, enabling unique functionalities such as tunable coupling[13, 14], non-reciprocity[15, 16, 17, 18], and quantum transduction[19]. As these device architectures continue toward nanoscale dimensions [20, 21], the complexity of electromagnetic interactions increases, particularly due to the emergence of indirect or unintended couplings between spatially separated components. Shared photonic modes or circuit environments often mediate these couplings and can lead to both detrimental crosstalk and useful long-range interactions [22, 23]. Understanding and controlling these mediated interactions is essential for designing scalable hybrid quantum devices that are also coherent and robust.

Material choice plays a crucial role in determining coherence, coupling strength, and scalability in hybrid quantum systems. Yttrium Iron Garnet (YIG), with its low damping and

long spin-wave coherence, has emerged as a leading candidate for magnonic quantum circuits [8, 14, 24, 25]. It enables strong and ultrastrong photon-magnon coupling in planar superconducting resonators, with recent studies[14] showing coupling-to-frequency ratios greater than 0.2. Likewise, lithographically patterned organic ferrimagnets have demonstrated cooperativity values exceeding 10^3 in the hybrid quantum modes [22], paving the way for their integration with superconducting qubit platforms and enabling circuit quantum electrodynamics (cQED) functionalities within magnon-based architectures. These advances highlight the growing maturity of hybrid photon-magnon coupled (PMC) systems. They underscore the need for deeper exploration into their nontrivial coupling dynamics, particularly those arising from indirect, photon-mediated interactions between spatially separated magnetic elements.

Unresolved theoretical and experimental challenges persist, particularly in understanding mediated coupling, the dependence of interaction strength on magnetization and spatial configuration, and the role of decoherence in nonlocal systems. Notably, Hyde *et al.* [26] observed that specific indirectly coupled modes can surpass their directly coupled counterparts in transmission amplitude, highlighting the need to reassess the conventional assumptions surrounding circuit isolation and crosstalk. As previous work suggests that cavity photons can act as effective mediators for long-distance coupling in hybrid architectures [27], we investigate this phenomenon using a hexagonal ring resonator as the coupling platform. It is planar and symmetric: well-suited for probing such mediated interactions and exploring their impact on the coherent dynamics of spatially separated magnetic elements.

This study demonstrates that photon-mediated interactions between a YIG film and a spatially separated permalloy film, integrated via the resonator, can give rise to measurable magnon-magnon coupling, even in the absence of significant direct dipolar overlap. Our choice of configuration is in contrast with systems often studied, such as YIG spheres[28, 29, 30, 31] and YIG/permalloy bilayers[32, 33, 34] that carry direct coupling. The observed coupling arises from the nonlocal electromagnetic modes of the resonator, which effectively bridge the two magnetic films. Importantly, the coupling strength is found to

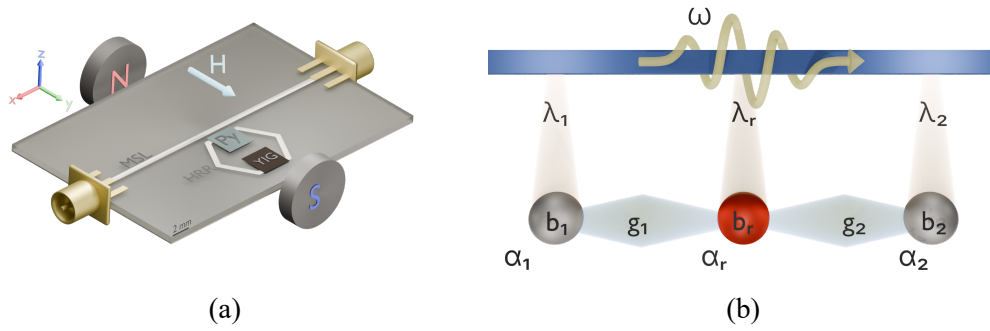


Figure 2.1: (a) The setup includes the two magnonic films placed on a pair of opposing sides of a copper hexagon (the photonic resonator). The ports are plugged into a vector network analyzer to both excite the microstrip line and also measure the transmission spectrum while a magnetic field is applied along the y -direction. (b) is the schematic of the model developed for this system. The spheres are bosonic oscillators with creation (annihilation) operator $\hat{b}_j^\dagger$ ($\hat{b}_j$). α_j denote intrinsic damping of the oscillators, and λ_j couple the oscillators to the traveling photon. The constants g_1 and g_2 denote the coupling of the magnonic cavities to the photonic cavity.

depend sensitively on the geometric configuration and electromagnetic response of circuit elements that are otherwise considered independent[26]. Notably, this mediated interaction persists even when the individual photon-magnon coupling lies well below the ultrastrong coupling threshold, emphasizing the relevance of indirect pathways in the overall system dynamics. These results suggest that conventional approaches to suppressing crosstalk in magnonic integrated circuits may need to be revised to account for long-range electromagnetic interactions. To model the system, we employ input–output theory[3, 35] instead of the conventional Schrieffer–Wolff transformation[36, 37], allowing us to extract (S_{21}) and directly compare theoretical predictions with simulation data. The design is intentionally simple and scalable: implemented on a millimeter scale, operable at room temperature. Despite its simplicity, the platform offers a promising testbed for investigating mediated coupling in accessible, low-cost quantum device architectures.

Chapter 3

Results and Discussion

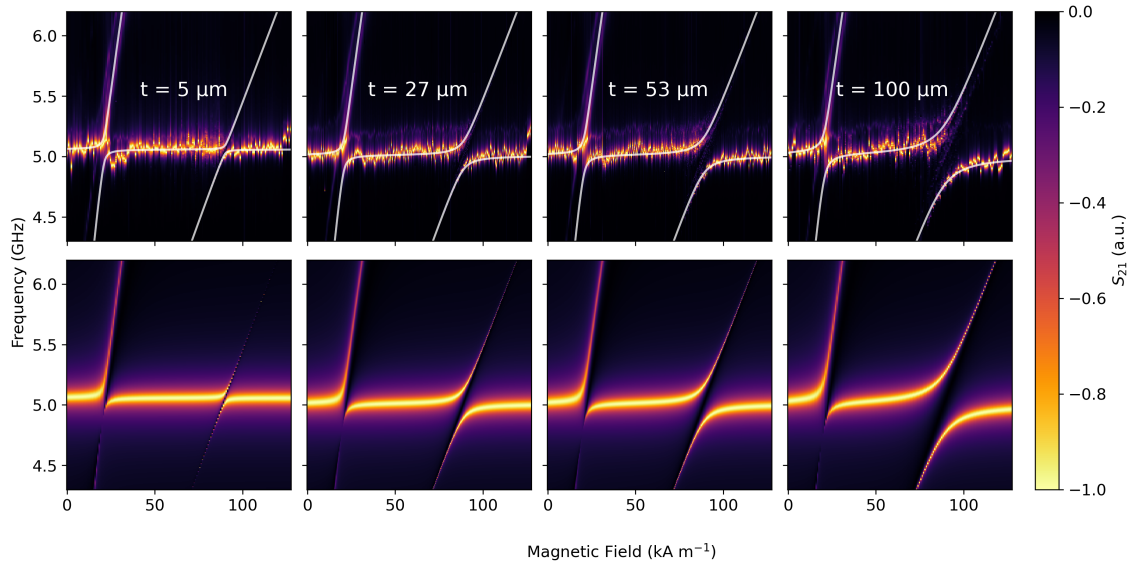


Figure 3.1: Observed (top row) and calculated (bottom row) S_{21} spectra of the system for corresponding selected thicknesses of YIG. The permalloy crossing is also seen to widen with a change in YIG thickness.

Choosing to exclude terms of direct coupling in the interaction Hamiltonian (4.3) was far from an arbitrary one. The configuration of the system is such that we observe minimal overlap between the two magnonic films on account of the close-to-planar thin films. This behavior is mirrored by the model, which accurately predicts the S_{21} spectrum in the absence of a direct coupling term.

The coupling constants predictably affect the extent of width of the avoided crossing for the corresponding magnon (identified by the coupling centers), with greater constants

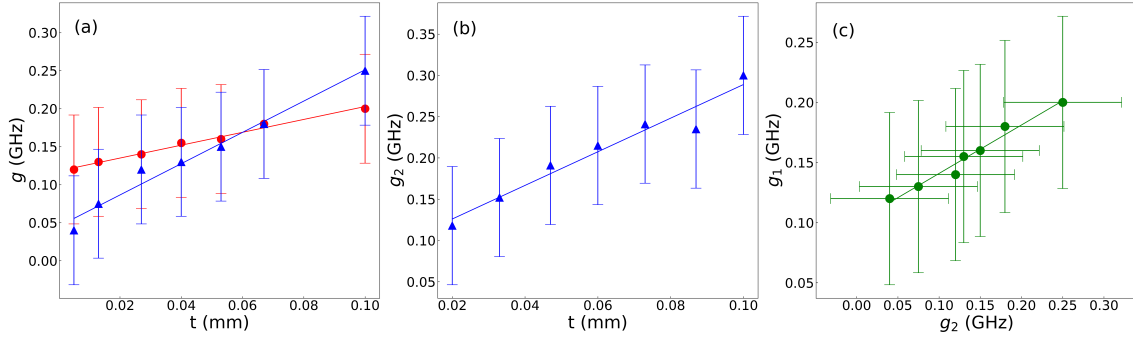


Figure 3.2: The blue crosses in (a) and (b) represent the coupling strengths of the YIG film to the resonator. The coupling strengths in (a) are for a system that does not include a permalloy film. Upon introduction of the permalloy in (b), the coupling strengths of the YIG have decreased by a fixed amount. The red dots in (b) mark g_1 (of permalloy), the coupling strengths of the permalloy to the resonator, and blue crosses are g_2 (of YIG), the coupling strengths of the YIG to the resonator. In (c), we plot the linear relationship between the two coupling strengths, $g_1(g_2)$.

leading to greater level repulsion.

To investigate the presence of indirect magnon–magnon coupling mediated by cavity photons, we performed a series of simulations by varying the thickness (t) of the YIG film (from $t = 5$ to $100 \mu\text{m}$) while keeping the properties of the permalloy film and the hexagonal microwave resonator unchanged. Increasing the YIG thickness enhances its magnetization, thereby modifying its magnonic response.

As shown in the top row of Figure 3.1, an increase in the mode splitting between the upper and lower branches (for both regions P1 and P2) of the magnon–photon hybrid modes is observed with thicker YIG films. Notably, this occurs when the permalloy film remains unaltered, indicating that the enhanced coupling is not due to changes in itself. We make a more direct observation of this in the plots in Figure 3.2, where we compare the coupling of the resonator–YIG in the absence of the permalloy film in Figure 3.2a and in its presence in Figure 3.2b (blue crosses). Both linear.

Our analysis shows that while the slope of the resonator–YIG coupling (i.e., g_2 vs t) remains unchanged from Figure 3.2b to Figure 3.2a, the intercept does drop slightly. The presence, then, of the permalloy film causes a decrease in resonator–YIG coupling independent of the thickness of the YIG film. Combined with the non-zero slope of the unvarying resonator–permalloy couple, the behavior provides strong evidence for photon–

mediated interaction between the YIG and permalloy magnon modes, a hallmark of indirect magnon–magnon coupling through the shared cavity photon field.

To validate these observations, we used a theoretical model to compute the transmission spectrum (S_{21}) under identical conditions. The bottom row of Figure 3.1 presents the calculated spectra, which closely match the corresponding observed results in the top row. The evolution of the spectra with increasing YIG thickness—from left to right—demonstrates consistent hybrid dynamics, affirming the model’s accuracy. Predictably, as YIG becomes thicker, its coupling to the resonator increases due to the higher magnetization. However, a notable and somewhat counterintuitive result is the simultaneous increase in the coupling strength of the permalloy mode, despite its parameters remaining fixed. This reveals that the YIG film exerts an indirect influence on the permalloy–resonator coupling, mediated by their mutual interaction with the cavity mode. We can ascribe to the resonator-permalloy coupling a dependence on the resonator-YIG coupling, $g_1(g_2)$, which we observe in this work to be linear (Figure 3.2c).

The observed spectra for the permalloy system appear noisy due to its metallic nature and higher damping parameters, which together broaden spectral features. Despite this, the systematic increase of the anti-crossing gap with YIG thickness is clearly discernible and reproduced by the theoretical model. Although the uncertainties are non-negligible in the estimation of coupling strength values, they do not obscure the linear dependence of coupling strength on YIG thickness; instead, the statistical analysis confirms systematic scaling consistent with theoretical predictions.

In this study, we have demonstrated and quantified the photon-mediated coupling between two spatially separated magnonic films, YIG and permalloy, integrated onto a planar hexagonal ring resonator. Our results reveal that even without direct dipolar interaction, a clear magnon-magnon coupling emerges, evidenced by the anticrossing in the transmission spectra. A key finding is the non-trivial enhancement of the permalloy-resonator coupling strength as the thickness of the YIG film increases, while the permalloy film itself remains unchanged. This observation strongly indicates that the resonator’s photonic

mode acts as a quantum bus, facilitating an indirect interaction pathway between the two magnonic systems.

The theoretical model developed using the input-output formalism accurately reproduces the observed transmission spectra across various YIG thicknesses. The model confirms a linear relationship between the coupling strengths of the YIG and permalloy films to the resonator, quantitatively capturing the indirect influence one magnon has on the other. These findings underscore the critical role of mediated interactions in hybrid magnonic systems, offering a new mechanism for controlling coupling strengths in scalable, on-chip architectures and paving the way for advanced quantum and microwave technologies.

While this study focuses on photon-mediated indirect coupling, exploring direct dipolar interactions and the transition between the direct and indirect coupling regimes offers a promising direction for future investigation that could provide deeper insight for understanding tunable magnonic networks.

This work demonstrates a magnon-controlled magnon–photon interface that serves as a fundamental building block for hybrid quantum devices. In the classical regime, this system can find utility as a parametric field detector, with more complex magnonic structures helping fingerprint magnetic fields. Possibly, given superconducting conditions and scaled-down magnetic films, it also achieves fine control of magnons, leading to a quantum sensor or even qubits.

Chapter 4

Methods and Calculations

4.1 Design and Numerical Modeling of the Hybrid Photon-Magnon System

In designing hybrid systems, it is essential to account for the various forms of hybridization and their collective influence on the system's dynamics. The hexagon is well-known for its tessellation: it is easily scaled and multiplied on a two-dimensional surface. However, our motivation to select a hexagonal structure was that it offers a superior balance of field localization and symmetry compared to circular or square rings. Such a geometry ensures stronger confinement of the microwave magnetic field within the trackwidth region and maximises overlap with the magnonic films enabling robust photon-magnon coupling[38], all while enhancing mode control and mitigating unwanted splitting. Together, these features yield significantly improved Q-factors, making the hexagonal resonator an optimal platform for the proposed hybrid system[39].

Previous work by Verma *et al.*[40] studied the mode profile of a similar set-up, comparing coupling strengths of two magnons for various positions on the hexagonal resonator. We therefore chose the 0° - 180° configuration to exploit maximal indirect magnon-magnon interaction. This allows us to isolate and analyze interaction pathways mediated indirectly through a standard bosonic mode, i.e., photons. The observed hybrid system features the

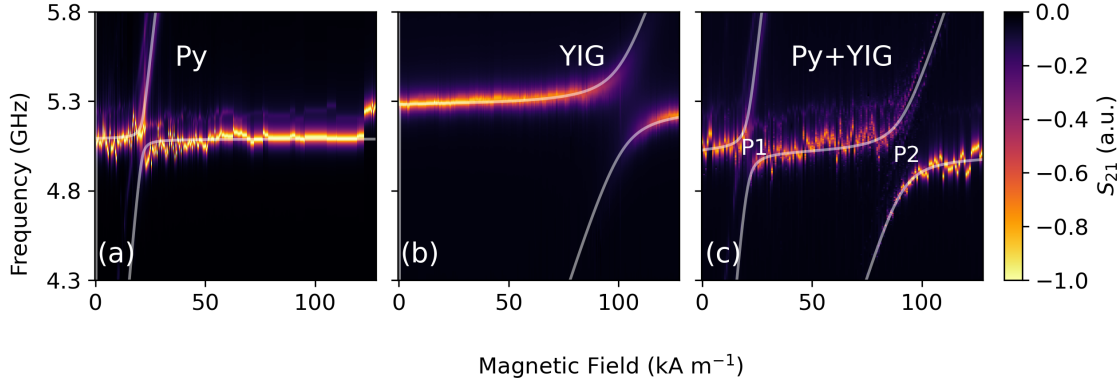


Figure 4.1: A comparison of the magnonic coupling strengths of the (a) permalloy, (b) YIG, and (c) both to the resonator in an indirectly coupled configuration. The overlaid white curves are the real part of the eigenvalues of (4.6) at each value of H and were used to fit the model.

resonator positioned near a microstrip line that is 0.57 mm wide and 30 mm long. It integrates with the two 3 mm $\times$ 3 mm magnetic films—YIG and permalloy—mounted on opposite arms of the resonator within a planar geometry, as illustrated in Figure 2.1a. When a microwave current propagates along the x -axis (see axes in Figure 2.1a) through the microstrip line (labelled ‘MSL’ in Figure 2.1a), it produces a transverse microwave magnetic field that excites the hexagonal resonator (labelled ‘HRR’). It functions as a parallel-LC circuit exhibiting quasi-static resonance. This resonant field, in turn, interacts with the magnetic YIG (black film in Figure 2.1a) and permalloy (of thickness 20 μm , metallic green film in the Figure) films placed on the resonator, inducing their respective magnon modes and enabling the probing of respective photon–magnon coupling dynamics. The high-frequency electromagnetic solver CST Studio Suite was employed to simulate individual hybrid configurations: resonator–permalloy, resonator–YIG, and the combined hybrid system resonator–YIG–permalloy.

To investigate photon-mediated magnon–magnon coupling, we obtained S_{21} as a function of microwave frequency (ω) under a static external magnetic field (H) applied along the y -axis at room temperature. Figures 4.1a, 4.1b, and 4.1c show the S_{21} power spectra on the ω – H plane for the hybrid resonator–permalloy, resonator–YIG, and resonator–YIG–permalloy, respectively. In this setup, the photon mode associated with the resonator behaves as a fixed-frequency resonance determined by the geometry of the LC structure. In

contrast, the magnon modes in YIG and permalloy follow the Kittel relation[41], resulting in a resonance that shifts with the applied magnetic field. The strength of photon-magnon coupling is inferred from the degree of avoided level crossing—an established indicator of coherent energy exchange—and is a particularly relevant parameter in applications such as quantum sensing and quantum memory buses[22].

As shown in Figure 4.1a, the permalloy–resonator system exhibits only weak coupling, which is expected due to the metallic nature of permalloy that introduces higher damping and noise. In our model, we observe a coupling constant of $g = 0.11$ corresponding to this interaction. In contrast, Figure 2b demonstrates that the YIG–resonator system exhibits strong coupling ($g = 0.25$), attributable to the low damping and high spin-wave coherence of the magnetic material involved. Outside the influence of the magnonic film’s resonance behavior, the photonic resonator returns to its natural frequency, observed in the rise of the horizontal line in Figure 4.1b.

Interestingly, in the combined system (Figure 4.1c), the permalloy mode displays a markedly enhanced coupling signature compared to its isolated behavior, while the YIG coupling appears slightly diminished. We note that fitted values are now $g_1 = 0.2$ (of permalloy) and $g_2 = 0.21$ (of YIG), a simultaneous enhancement of permalloy-resonator coupling (seen at crossing marked P1 in Figure 4.1c) and reduction of YIG-resonator coupling (crossing P2). This suggests a redistribution of interaction strength facilitated by the resonator’s photon mode, emphasizing the role of photon mediation in enabling effective magnon–magnon interactions between spatially separated magnetic elements.

4.2 Theoretical Formalism for Photon Mediated Magnon-Magnon Coupling

We developed a microscopic model based on quantum mechanical principles to gain deeper insight into the numerically observed photon-mediated magnon–magnon coupling and quantitatively estimate the corresponding interaction strength. This approach provides

a robust framework for describing the coupled dynamics of the system, effectively capturing both direct and indirect interactions. It is readily extended to systems involving multiple coupled quantum harmonic oscillators.

Figure 2.1b attempts to visually illuminate the nature of this model. Three oscillators named 1, 2, and r , each with its own internal damping and a coupling to travelling photons, interact with one another. While oscillators 1 and 2 interact with oscillator r directly, they are not allowed to interact with one another.

The system Hamiltonian for such a model is best understood as a sum of three parts, $\hat{H} = \hat{H}_o + \hat{H}_{\text{msl}} + \hat{H}_{\text{int}}$. The natural Hamiltonian $\hat{H}_o$, is the familiar sum of quantum harmonic oscillators,

$$\hat{H}_o/\hbar = \sum_{j \in \{r, 1, 2\}} \tilde{\omega}_j \hat{b}_j^\dagger \hat{b}_j \quad (4.1)$$

where $\tilde{\omega}_j = \omega_j - i\alpha_j$ refers to the damped natural frequency of the oscillator for a resonant frequency ω_j and an intrinsic damping constant α_j . For the index j , we use r to denote the photonic resonator and 1 and 2 to denote the permalloy and YIG magnons, respectively.

We then provide a driving field through a micro stripline, which is assigned the annihilation (creation) operator $\hat{p}_k$ ($\hat{p}_k^\dagger$) for the k th mode:

$$\hat{H}_{\text{msl}}/\hbar = \int \omega_k \hat{p}_k^\dagger \hat{p}_k dk + \sum_{j \in \{r, 1, 2\}} \int \lambda_j \left(\hat{b}_j^\dagger + \hat{b}_j \right) \left(\hat{p}_k^\dagger + \hat{p}_k \right) dk \quad (4.2)$$

where $\hat{b}_j$ ($\hat{b}_j^\dagger$) are the bosonic annihilation (creation) operators. We introduce λ_j to denote the coupling of the oscillators to the stripline.

The final term in the Hamiltonian is the mutual interaction among all the resonators, photonic and magnonic. The coupling constants g_j that populate $\hat{H}_{\text{int}}$ are of primary interest to us. This part of the Hamiltonian is given by

$$\hat{H}_{\text{int}}/\hbar = g_1 \hat{b}_1^\dagger \hat{b}_r + g_2 \hat{b}_2^\dagger \hat{b}_r + \text{h.c.} \quad (4.3)$$

and captures the crosstalk between the resonator modes, leading to the observed normal

anticrossing in Figure 4.1. The normal anticrossing behaviour is due to real coupling constants in the Hamiltonian [42, 43].

The equations of motion for the three oscillators are readily calculated by the standard Heisenberg-Langevin formalism [44, 35]. Following a method similar to that followed by Srivastava *et al.* [45] we set $\beta_i = 2\pi\lambda_i^2$, the S_{21} parameter for such a system, given by $S_{21} = \hat{p}_{\text{out}}/\hat{p}_{\text{in}} - 1$ is

$$S_{21} = \frac{2}{i} \sum_j \sqrt{\beta_j} \frac{\hat{b}_j(\omega)}{\hat{p}_{\text{in}}(\omega)} \quad (4.4)$$

which, rewritten in the form of a matrix

$$S_{21} = B^T \mathcal{M}^{-1} B \quad (4.5)$$

where

$$B = \sqrt{2} \begin{bmatrix} \sqrt{\beta_1} \\ \sqrt{\beta_r} \\ \sqrt{\beta_2} \end{bmatrix},$$

and $\mathcal{M} = i(\omega\mathcal{I} - \hat{H}_{\text{coupling}})$, where $\mathcal{I}$ is the 3×3 identity matrix and $\hat{H}_{\text{coupling}}$ is the effective coupling Hamiltonian, that solves the equations of motion for the mode operators, given by

$$\hat{H}_{\text{coupling}} = \begin{bmatrix} \tilde{\omega}'_1 & g_1 - i\sqrt{\beta_1\beta_r} & -i\sqrt{\beta_1\beta_2} \\ g_1 - i\sqrt{\beta_1\beta_r} & \tilde{\omega}'_r & g_2 - i\sqrt{\beta_r\beta_2} \\ -i\sqrt{\beta_1\beta_2} & g_2 - i\sqrt{\beta_r\beta_2} & \tilde{\omega}'_2 \end{bmatrix}. \quad (4.6)$$

The terms on the diagonal include intrinsic and extrinsic damping terms, in the form $\tilde{\omega}'_j = \tilde{\omega}_j - i\beta_j$ [45]. Since we are ignoring terms of direct magnon coupling, there are no coupling constants on the edges of the off-diagonal. The parameters g_1 and g_2 can be obtained by fitting the model to observations at each H .

4.3 Spatial distributions of microwave magnetic field distribution at the resonance peak of the hexagonal ring resonator (HRR)

We calculated the spatial distributions of electric and microwave magnetic field for the hybrid magnon–photon–magnon system under an applied bias field of 1100 Oe (~ 87 kA/m), as shown in Figure 4.2. Panel (a) shows the S-parameter spectrum of the system, where the hybridized mode corresponding to the resonance valley at 5.15 GHz. Panel (b) displays the electric field distribution, while panel (c) illustrates the corresponding microwave magnetic (h -field) distribution at this hybrid resonance. The field profiles clearly demonstrate the simultaneous excitation of the photonic cavity mode and the magnonic mode in the Yttrium Iron Garnet (YIG) film, confirming coherent photon–magnon hybridization. It is noteworthy that only one of the two magnonic modes is excited in this configuration, which arises from the orientation of the bias magnetic field and the probe frequency selectively matching the resonance condition of the lower YIG film.

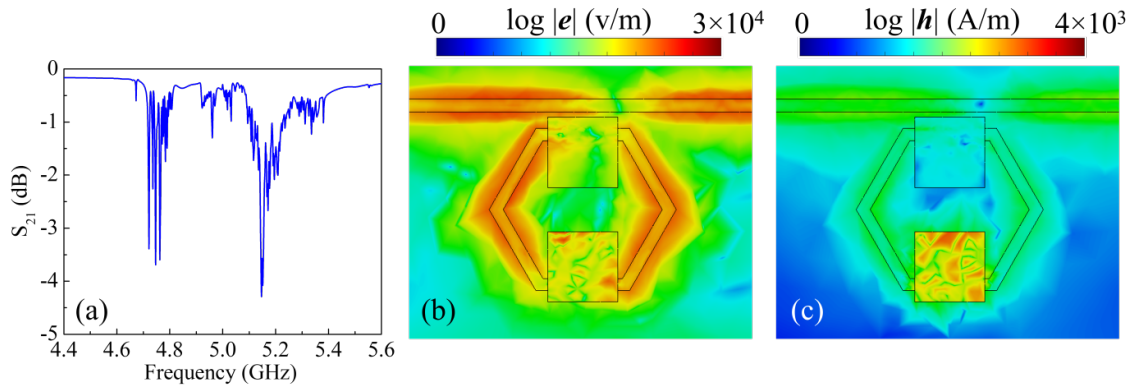


Figure 4.2: Simulated transmission spectra and field distributions of the hybrid magnon–photon–magnon system at 1100 Oe (~ 87 kA/m). (a) S-parameter spectrum showing the hybridized resonance (b) electric field distribution, and (c) magnetic (h -field) distribution at this resonance. The colors represent the strength of the electric and magnetic field in the respective field profile.

Figure 4.3 shows the simulated microwave magnetic field distribution around the hexagonal ring resonator at its resonant frequency, obtained from CST Microwave Studio. The color map illustrates the spatial localization and intensity of the magnetic field, showing

strong confinement and enhanced field concentration near the HRR edges.

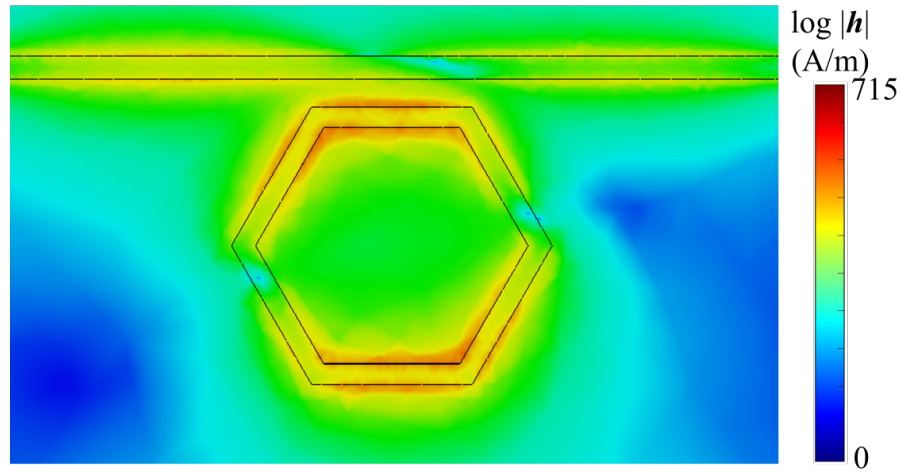


Figure 4.3: Spatial distributions of microwave magnetic field (h) distribution of HRR at resonance frequency $f = 5.29$ GHz. The colors represent the strength of the magnetic field.

Chapter 5

Conclusion and Outlook

In this study, we have demonstrated and quantified the photon-mediated coupling between two spatially separated magnonic films, YIG and permalloy, integrated onto a planar hexagonal ring resonator. Our results reveal that even without direct dipolar interaction, a clear magnon-magnon coupling emerges, evidenced by the anticrossing in the transmission spectra. A key finding is the non-trivial enhancement of the permalloy-resonator coupling strength as the thickness of the YIG film increases, while the permalloy film itself remains unchanged. This observation strongly indicates that the resonator's photonic mode acts as a quantum bus, facilitating an indirect interaction pathway between the two magnonic systems.

The theoretical model developed using the input-output formalism accurately reproduces the observed transmission spectra across various YIG thicknesses. The model confirms a linear relationship between the coupling strengths of the YIG and permalloy films to the resonator, quantitatively capturing the indirect influence one magnon has on the other. These findings underscore the critical role of mediated interactions in hybrid magnonic systems, offering a new mechanism for controlling coupling strengths in scalable, on-chip architectures and paving the way for advanced quantum and microwave technologies.

The implications spread across several fronts. On the sensing side, exceptional points[[46](#), [47](#)] in coupled magnon-photon systems have enabled ultrasensitive detection schemes.

Quantum information applications followed naturally, with proposals and demonstrations spanning memory architectures, quantum transduction, and optical-to-microwave frequency conversion — the last of which is particularly relevant for interfacing superconducting qubits with optical networks. Beyond applications, cavity magnonics has also served as a clean testbed for fundamental physics, particularly dissipative coupling phenomena and the behavior of exceptional points in open quantum systems.

While this study focuses on photon-mediated indirect coupling, exploring direct dipolar interactions and the transition between the direct and indirect coupling regimes offers a promising direction for future investigation that could provide deeper insight for understanding tunable magnonic networks.

This work demonstrates a magnon-controlled magnon–photon interface that serves as a fundamental building block for hybrid quantum devices. In the classical regime, this system can find utility as a parametric field detector, with more complex magnonic structures helping fingerprint magnetic fields. Possibly, given superconducting conditions and scaled-down magnetic films, it also achieves fine control of magnons, leading to a quantum sensor or even qubits.

Appendices

Appendix A

Derivation of transmission parameter

$$S_{21}$$

The following is a detailed derivation of the transmission parameter S_{21} , using the input-output formalism[3, 45].

The Hamiltonian of the system was asserted in Section 4.2 to exist in three parts: (4.1), (4.2), and (4.3). Plugging these into $\hat{H} = \hat{H}_o + \hat{H}_{\text{msl}} + \hat{H}_{\text{int}}$ and making the rotating wave approximation we obtain

$$\hat{H}/\hbar = \sum_{j \in \{r, 1, 2\}} \left[\tilde{\omega}_j \hat{b}_j^\dagger \hat{b}_j + \int_{-\infty}^{\infty} \lambda_j (\hat{b}_j^\dagger \hat{p}_k + \hat{p}_k^\dagger \hat{b}_j) dk \right] + [g_1 \hat{b}_1^\dagger \hat{b}_r + g_2 \hat{b}_2^\dagger \hat{b}_r + h.c.] + \int_{-\infty}^{\infty} \omega_k \hat{p}_k^\dagger \hat{p}_k dk \quad (\text{A.1})$$

The dynamics of the system can be described by the Heisenberg-Langevin equations of motion, which are derived from the Hamiltonian and the fundamental commutation relations for bosonic operators that have the commutation relations $[\hat{p}_{k'}, \hat{p}_k] = \delta(k' - k)$ and $[\hat{b}_i^\dagger, \hat{b}_j] = \delta_{i,j}$.

The solutions to the Heisenberg equations of motion for $\hat{p}_k(t)$ and $\hat{p}_k^\dagger(t)$ for some $t_0 < t$

are the following:

$$\hat{p}_k(t) = e^{-i\omega_k(t-t_0)}\hat{p}_k(t_0) - i \sum_j \lambda_j \int_{t_0}^t \hat{b}_j(t') e^{-i\omega_k(t-t')} dt' \quad (\text{A.2a})$$

$$\hat{p}_k^\dagger(t) = e^{-i\omega_k(t-t_0)}\hat{p}_k^\dagger(t_0) - i \sum_j \lambda_j \int_{t_0}^t \hat{b}_j^\dagger(t') e^{-i\omega_k(t-t')} dt' \quad (\text{A.2b})$$

Whereas, when the solution is sought until some $t_1 > t$:

$$\hat{p}_k(t) = e^{-i\omega_k(t-t_1)}\hat{p}_k(t_1) + i \sum_j \lambda_j \int_t^{t_1} \hat{b}_j(t') e^{-i\omega_k(t-t')} dt' \quad (\text{A.3a})$$

$$\hat{p}_k^\dagger(t) = e^{-i\omega_k(t-t_1)}\hat{p}_k^\dagger(t_1) + i \sum_j \lambda_j \int_t^{t_1} \hat{b}_j^\dagger(t') e^{-i\omega_k(t-t')} dt' \quad (\text{A.3b})$$

We pull the λ_j out of the integral under the assumption that the transfer of energy between the operators $\hat{p}_k$ and $\hat{b}_j$ is a Markov process. For $\hat{b}_j$:

$$\dot{\hat{b}}_{1,2}(t) = -i\tilde{\omega}_{1,2}\hat{b}_{1,2}(t) - i\lambda_{1,2} \int \hat{p}_k(t) dk - ig_{1,2}\hat{b}_r(t) \quad (\text{A.4a})$$

$$\dot{\hat{b}}_r(t) = -i\tilde{\omega}_r\hat{b}_r(t) - i\lambda_r \int \hat{p}_k(t) dk - i(g_1^*\hat{b}_1 + g_2^*\hat{b}_2)(t). \quad (\text{A.4b})$$

We will then define $\hat{p}_{\text{in}}(t)$ and $\hat{p}_{\text{out}}(t)$ as

$$\hat{p}_{\text{in}}(t) = \frac{1}{\sqrt{2\pi}} \int e^{-i\omega_k(t-t_0)} \hat{p}_k(t_0) dk \quad (\text{A.5a})$$

$$\hat{p}_{\text{out}}(t) = \frac{1}{\sqrt{2\pi}} \int e^{-i\omega_k(t-t_1)} \hat{p}_k(t_1) dk. \quad (\text{A.5b})$$

We can now define the decay rate $\beta_j = 2\pi\lambda_j^2$. This allows us to further simplify the equations of motion by plugging the equations of the input or output modes (A.5) into

(A.4) to yield

$$\dot{\hat{b}}_j(t) = -i\tilde{\omega}_j\hat{b}_j(t) - i\sqrt{\beta_j}\hat{p}_{\text{in}}(t) - \sum_{n \in \{r,1,2\}} \sqrt{\beta_j\beta_n}\hat{b}_n(t) - i\hat{\phi}_j(t) \quad (\text{A.6a})$$

$$\dot{\hat{b}}_j(t) = -i\tilde{\omega}_j\hat{b}_j(t) - i\sqrt{\beta_j}\hat{p}_{\text{out}}(t) + \sum_{n \in \{r,1,2\}} \sqrt{\beta_j\beta_n}\hat{b}_n(t) - i\hat{\phi}_j(t). \quad (\text{A.6b})$$

where $\hat{\phi}_j(t)$ is defined as

$$\hat{\phi}_j(t) = \begin{cases} g_1^*\hat{b}_1(t) + g_2^*\hat{b}_2(t) & ; j = r \\ g_j\hat{b}_r(t) & ; j = 1, 2 \end{cases}. \quad (\text{A.7})$$

Taking the Fourier transform of these expanded equations gives

$$i(\tilde{\omega}_j \pm \omega)\hat{b}_j(\omega) + i\sqrt{\beta_j}\hat{p}_{\text{in}}(\omega) + \sum_n \sqrt{\beta_j\beta_n}\hat{b}_n(\omega) + i\hat{\phi}_j(\omega) = 0 \quad (\text{A.8a})$$

$$i(\tilde{\omega}_j \pm \omega)\hat{b}_j(\omega) + i\sqrt{\beta_j}\hat{p}_{\text{out}}(\omega) - \sum_n \sqrt{\beta_j\beta_n}\hat{b}_n(\omega) + i\hat{\phi}_j(\omega) = 0. \quad (\text{A.8b})$$

Finally, by defining the S_{21} parameter as $S_{21} = \hat{p}_{\text{out}}/\hat{p}_{\text{in}} - 1$ and substituting the derived expressions, we arrive at the final result:

$$S_{21}(\omega) = \frac{2}{i} \sum_n \sqrt{\beta_n} \frac{\hat{b}_n(\omega)}{\hat{p}_{\text{in}}(\omega)}, \quad (\text{A.9})$$

which matches Equation 4.4 precisely.

Appendix B

Experimental Setup

B.1 Hardware

The equipment that ran the experiments was a combination of a vector network analyzer (VNA) and a magnet. The VNA was used to measure the transmission spectra of the system, while the magnet provided the DC external magnetic field necessary to tune the magnon modes in the YIG and permalloy films perpendicular to the MSL. The VNA was connected to the microstrip line (MSL) of the sample, allowing us to probe the microwave response of the system as a function of frequency and applied magnetic field by the SMA connectors visualized as the brass connects on the ends of the substrate in Fig. [B.1](#).

Parameter	Specification
Frequency Range	1 MHz to 18 GHz (overrange to 20 GHz); extends down to 100 kHz with the ZNLE-B100 option
Number of Ports	2 ports (for bidirectional measurements)
Dynamic Range	Up to 120 dB (typical)
Measurement Bandwidth	1 Hz to 500 kHz
Measurement Speed	8.7 ms for 401 points (100 kHz IFBW, 200 MHz span, correction off)

Table B.1: Specifications for the Rohde & Schwarz ZNLE18 Vector Network Analyzer.

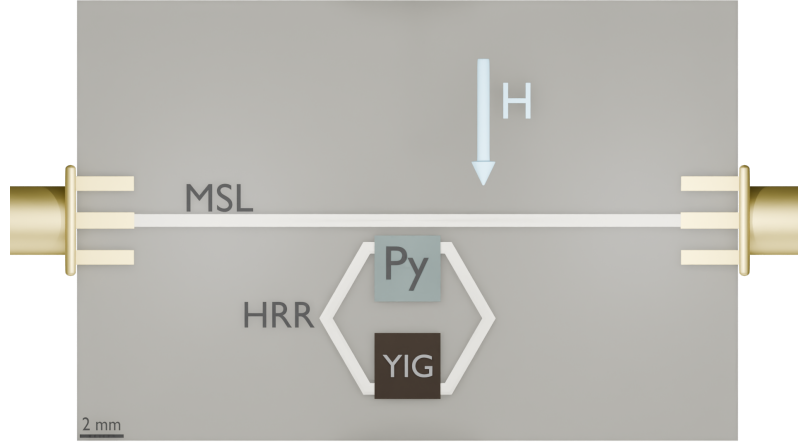


Figure B.1: Top-down view of the sample, showing the hexagonal ring resonator (HRR) and the microstrip line (MSL) with SMA connectors. The YIG and permalloy films are mounted on opposite arms of the HRR, allowing for the study of photon-mediated magnon-magnon coupling. The magnetic field is applied perpendicular to the MSL, as denoted by the blue H vector.



Figure B.2: Photograph of the experimental setup, showing the Rhode&Schwartz ZNLE18 vector network analyzer (left) connected to the sample via SMA connectors, and the Holmarc EM3000S electromagnet (right) used to apply the external magnetic field. The sample is placed in the gap of the magnet, allowing for precise control of the magnetic field while measuring the transmission spectra.

Parameter	Specification
Maximum Magnetic Field	2.1 T at a 5 mm pole gap
Magnetic Field Uniformity	Uniform over 5 cm ³
Field Stability	< 0.005% (for durations $\geq$ 8 hours)
Cooling System	Water-cooled (motor pump based, minimum 7.6 L/min flow rate)
Power Supply Output	Constant current bipolar, up to 70V Max, 4 Amps continuously variable
Remote Monitoring/Interface	USB port

Table B.2: Specifications for the Holmarc EM3000S Electromagnet.

B.2 Software

B.2.1 Drivers

The EM3000S came with proprietary control software, which sent non-SCPI signals to the electromagnet to control the magnetic field. An attempt to send SCPI signals to the electromagnet was unsuccessful, with the magnet not responding to any commands sent through the USB port. To solve this, we used a port sniffer to sniff (our own machine's) USB traffic while using the proprietary software to control the magnet, and reverse engineered the commands sent to the magnet to control it through a custom Python script. The magnet takes hex digits that perform functions such as reading and writing the electromagnet's current, or sending a stop signal which appears to also perform a degaussing on the magnet. For writing a current, the hex digits needed to be fitted to a cubic function to output current with linear control. Around zero however, the function has jumps corresponding to $\tilde{10}\Delta I$ where ΔI is the current resolution.

The VNA was controlled through the standard SCPI commands sent through the LAN port, which allowed us to automate the measurement of S_{21} as a function of frequency and magnetic field, wait for the magnetic field to change, and make the next measurement. Thus automating a full sweep of the magnetic field and frequency space.

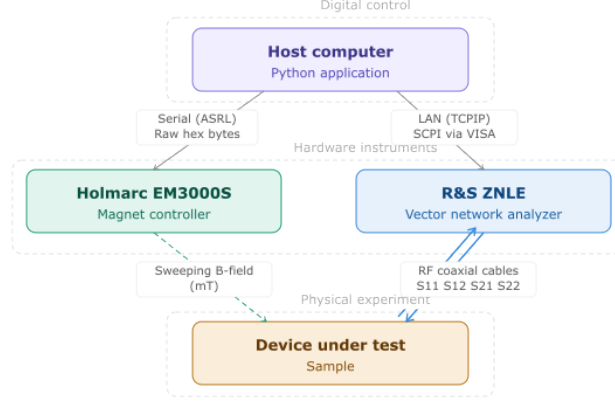


Figure B.3: Overview of the codebase architecture, showing the main components and their interactions.

B.2.2 Stack

The codebase (available on [GitHub](#)) was written in Python, using the PyVISA library to control the VNA and the electromagnet, and aims to see continued improvements in the future. Readme files within the repository go into greater detail about the codebase, its usage and development.

B.3 Data Analysis

The data obtained from the VNA was in the form of S_{21} as a function of frequency and magnetic field. We used Python to analyze this data, fitting the observed transmission spectra to the theoretical model derived in Section 4.2 to extract the coupling constants g_1 and g_2 for the permalloy and YIG films, respectively. The fitting process involved minimizing the difference between the observed S_{21} spectra and the theoretical predictions by adjusting the coupling constants, allowing us to quantify the strength of photon-mediated magnon-magnon coupling in our system.

References

- [1] Yi Li et al. “Hybrid magnonics: Physics, circuits, and applications for coherent information processing”. In: *Journal of Applied Physics* 128.13 (Oct. 2020), p. 130902. ISSN: 0021-8979. DOI: [10.1063/5.0020277](https://doi.org/10.1063/5.0020277). eprint: https://pubs.aip.org/aip/jap/article-pdf/doi/10.1063/5.0020277/20015277/130902_1_5.0020277.pdf. URL: <https://doi.org/10.1063/5.0020277>.
- [2] Ö. O. Soykal and M. E. Flatté. “Strong Field Interactions between a Nanomagnet and a Photonic Cavity”. In: *Phys. Rev. Lett.* 104 (7 Feb. 2010), p. 077202. DOI: [10.1103/PhysRevLett.104.077202](https://link.aps.org/doi/10.1103/PhysRevLett.104.077202). URL: <https://link.aps.org/doi/10.1103/PhysRevLett.104.077202>.
- [3] D.F. Walls and Gerard J. Milburn. *Quantum Optics*. Springer Science & Business Media, Dec. 2007.
- [4] M. G. Raizen et al. “Normal-mode splitting and linewidth averaging for two-state atoms in an optical cavity”. In: *Phys. Rev. Lett.* 63 (3 July 1989), pp. 240–243. DOI: [10.1103/PhysRevLett.63.240](https://link.aps.org/doi/10.1103/PhysRevLett.63.240). URL: <https://link.aps.org/doi/10.1103/PhysRevLett.63.240>.
- [5] Hans Huebl et al. “High Cooperativity in Coupled Microwave Resonator Ferrimagnetic Insulator Hybrids”. In: *Phys. Rev. Lett.* 111 (12 Sept. 2013), p. 127003. DOI: [10.1103/PhysRevLett.111.127003](https://link.aps.org/doi/10.1103/PhysRevLett.111.127003). URL: <https://link.aps.org/doi/10.1103/PhysRevLett.111.127003>.
- [6] Xufeng Zhang et al. “Strongly Coupled Magnons and Cavity Microwave Photons”. In: *Phys. Rev. Lett.* 113 (15 Oct. 2014), p. 156401. DOI: [10.1103/PhysRevLett.113.156401](https://doi.org/10.1103/PhysRevLett.113.156401).

- 113.156401. URL: <https://link.aps.org/doi/10.1103/PhysRevLett.113.156401>.
- [7] Fizaan Khan et al. “Dynamic Control of Photon–Magnon Interactions via Secondary Magnon Excitation”. In: *physica status solidi (RRL) – Rapid Research Letters* n/a.n/a (), e202500353. DOI: <https://doi.org/10.1002/pssr.202500353>. eprint: <https://onlinelibrary.wiley.com/doi/pdf/10.1002/pssr.202500353>. URL: <https://onlinelibrary.wiley.com/doi/abs/10.1002/pssr.202500353>.
- [8] Dinesh Wagle et al. “Controlling magnon-photon coupling in a planar geometry”. In: *Journal of Physics: Materials* 7.2 (Feb. 2024), p. 025005. ISSN: 2515-7639. DOI: [10.1088/2515-7639/ad2984](https://doi.org/10.1088/2515-7639/ad2984). URL: <http://dx.doi.org/10.1088/2515-7639/ad2984>.
- [9] Chengsong Zhao et al. “Nonreciprocal amplification in a cavity magnonics system”. In: *Phys. Rev. A* 105 (2 Feb. 2022), p. 023709. DOI: [10.1103/PhysRevA.105.023709](https://doi.org/10.1103/PhysRevA.105.023709). URL: <https://link.aps.org/doi/10.1103/PhysRevA.105.023709>.
- [10] Na Zhu et al. “Magnon-photon strong coupling for tunable microwave circulators”. In: *Phys. Rev. A* 101 (4 Apr. 2020), p. 043842. DOI: [10.1103/PhysRevA.101.043842](https://doi.org/10.1103/PhysRevA.101.043842). URL: <https://link.aps.org/doi/10.1103/PhysRevA.101.043842>.
- [11] Chi Zhang et al. “Nonreciprocal multimode and indirect couplings in cavity magnonics”. In: *Phys. Rev. B* 103 (18 May 2021), p. 184427. DOI: [10.1103/PhysRevB.103.184427](https://doi.org/10.1103/PhysRevB.103.184427). URL: <https://link.aps.org/doi/10.1103/PhysRevB.103.184427>.
- [12] Zhiyuan Fan et al. “Nonreciprocal entanglement in cavity magnomechanics exploiting chiral cavity-magnon coupling”. In: *Fundamental Research* (2025). ISSN: 2667-3258. DOI: <https://doi.org/10.1016/j.fmre.2025.02.012>. URL: <https://www.sciencedirect.com/science/article/pii/S2667325825000950>.
- [13] Yi Li et al. “Probing intrinsic magnon bandgap in a layered hybrid perovskite antiferromagnet by a superconducting resonator”. In: *Phys. Rev. Res.* 5 (4 Oct. 2023),

- p. 043031. DOI: [10.1103/PhysRevResearch.5.043031](https://doi.org/10.1103/PhysRevResearch.5.043031). URL: <https://link.aps.org/doi/10.1103/PhysRevResearch.5.043031>.
- [14] Alberto Ghirri et al. “Ultrastrong Magnon-Photon Coupling Achieved by Magnetic Films in Contact with Superconducting Resonators”. In: *Phys. Rev. Appl.* 20 (2 Aug. 2023), p. 024039. DOI: [10.1103/PhysRevApplied.20.024039](https://doi.org/10.1103/PhysRevApplied.20.024039). URL: <https://link.aps.org/doi/10.1103/PhysRevApplied.20.024039>.
- [15] Junyoung Kim et al. “Magnetic-field controlled on-off switchable non-reciprocal negative refractive index in non-Hermitian photon-magnon hybrid systems”. In: *Nature Communications* 15.1 (Oct. 2024), p. 9014. ISSN: 2041-1723. DOI: [10.1038/s41467-024-53328-9](https://doi.org/10.1038/s41467-024-53328-9). URL: <https://doi.org/10.1038/s41467-024-53328-9>.
- [16] Charlie W. F. Freeman et al. “Tunable Ultrastrong Magnon–Magnon Coupling Approaching the Deep-Strong Regime in a van der Waals Antiferromagnet”. In: *ACS Nano* 19.16 (2025). PMID: 40244900, pp. 16024–16031. DOI: [10.1021/acsnano.5c02576](https://doi.org/10.1021/acsnano.5c02576). eprint: <https://doi.org/10.1021/acsnano.5c02576>. URL: <https://doi.org/10.1021/acsnano.5c02576>.
- [17] M. Cosset-Chéneau et al. “Nonreciprocal transport of thermally generated magnons”. In: *Phys. Rev. B* 110 (21 Dec. 2024), p. 214418. DOI: [10.1103/PhysRevB.110.214418](https://doi.org/10.1103/PhysRevB.110.214418). URL: <https://link.aps.org/doi/10.1103/PhysRevB.110.214418>.
- [18] Subhadeep Chakraborty and Camelia Das. “Nonreciprocal magnon-photon-phonon entanglement in cavity magnomechanics”. In: *Phys. Rev. A* 108 (6 Dec. 2023), p. 063704. DOI: [10.1103/PhysRevA.108.063704](https://doi.org/10.1103/PhysRevA.108.063704). URL: <https://link.aps.org/doi/10.1103/PhysRevA.108.063704>.
- [19] Roson Nongthombam, Pooja Kumari Gupta, and Amarendra K. Sarma. “Quantum transduction of a superconducting qubit in an electro-optomechanical and an electro-optomagnonical system”. In: *Phys. Rev. A* 108 (4 Oct. 2023), p. 043501. DOI: [10.1103/PhysRevA.108.043501](https://doi.org/10.1103/PhysRevA.108.043501). URL: <https://link.aps.org/doi/10.1103/PhysRevA.108.043501>.

-
- [20] Justin T. Hou and Luqiao Liu. “Strong Coupling between Microwave Photons and Nanomagnet Magnons”. In: *Phys. Rev. Lett.* 123 (10 Sept. 2019), p. 107702. DOI: [10.1103/PhysRevLett.123.107702](https://doi.org/10.1103/PhysRevLett.123.107702). URL: <https://link.aps.org/doi/10.1103/PhysRevLett.123.107702>.
- [21] Yi Li et al. “Strong Coupling between Magnons and Microwave Photons in On-Chip Ferromagnet-Superconductor Thin-Film Devices”. In: *Phys. Rev. Lett.* 123 (10 Sept. 2019), p. 107701. DOI: [10.1103/PhysRevLett.123.107701](https://doi.org/10.1103/PhysRevLett.123.107701). URL: <https://link.aps.org/doi/10.1103/PhysRevLett.123.107701>.
- [22] Qin Xu et al. “Strong Photon-Magnon Coupling Using a Lithographically Defined Organic Ferrimagnet”. In: *Advanced Science* 11.14 (2024), p. 2310032. DOI: <https://doi.org/10.1002/advs.202310032>. URL: <https://advanced.onlinelibrary.wiley.com/doi/abs/10.1002/advs.202310032>.
- [23] Haechan Jeon et al. “Anomalous coherent and dissipative coupling in dual photon-magnon hybrid resonators”. In: *Scientific Reports* 14.1 (June 2024), p. 13581. ISSN: 2045-2322. DOI: [10.1038/s41598-024-64315-x](https://doi.org/10.1038/s41598-024-64315-x). URL: <https://doi.org/10.1038/s41598-024-64315-x>.
- [24] Biswanath Bhoi et al. “Robust magnon-photon coupling in a planar-geometry hybrid of inverted split-ring resonator and YIG film”. In: *Scientific Reports* 7.1 (Sept. 2017), p. 11930. ISSN: 2045-2322. DOI: [10.1038/s41598-017-12215-8](https://doi.org/10.1038/s41598-017-12215-8). URL: <https://doi.org/10.1038/s41598-017-12215-8>.
- [25] Paolo Andrich et al. “Long-range spin wave mediated control of defect qubits in nanodiamonds”. In: *npj Quantum Information* 3.1 (July 2017). ISSN: 2056-6387. DOI: [10.1038/s41534-017-0029-z](https://doi.org/10.1038/s41534-017-0029-z). URL: <http://dx.doi.org/10.1038/s41534-017-0029-z>.
- [26] Paul Hyde et al. “Indirect coupling between two cavity modes via ferromagnetic resonance”. In: *Applied Physics Letters* 109.15 (Oct. 2016), p. 152405. ISSN: 0003-6951. DOI: [10.1063/1.4964602](https://doi.org/10.1063/1.4964602). URL: <https://doi.org/10.1063/1.4964602>.

- [27] Vahram L. Grigoryan and Ke Xia. “Cavity-mediated dissipative spin-spin coupling”. In: *Phys. Rev. B* 100 (1 July 2019), p. 014415. DOI: [10.1103/PhysRevB.100.014415](https://doi.org/10.1103/PhysRevB.100.014415). URL: <https://link.aps.org/doi/10.1103/PhysRevB.100.014415>.
- [28] R. G. E. Morris et al. “Strong coupling of magnons in a YIG sphere to photons in a planar superconducting resonator in the quantum limit”. In: *Scientific Reports* 7.1 (Sept. 2017), p. 11511. ISSN: 2045-2322. DOI: [10.1038/s41598-017-11835-4](https://doi.org/10.1038/s41598-017-11835-4). URL: <https://doi.org/10.1038/s41598-017-11835-4>.
- [29] Xufeng Zhang et al. “Cavity magnomechanics”. In: *Science Advances* 2.3 (2016), e1501286. DOI: [10.1126/sciadv.1501286](https://doi.org/10.1126/sciadv.1501286). eprint: <https://www.science.org/doi/pdf/10.1126/sciadv.1501286>. URL: <https://www.science.org/doi/abs/10.1126/sciadv.1501286>.
- [30] J. Bourhill et al. “Ultrahigh cooperativity interactions between magnons and resonant photons in a YIG sphere”. In: *Phys. Rev. B* 93 (14 Apr. 2016), p. 144420. DOI: [10.1103/PhysRevB.93.144420](https://doi.org/10.1103/PhysRevB.93.144420). URL: <https://link.aps.org/doi/10.1103/PhysRevB.93.144420>.
- [31] Dilawaiz, Shahid Qamar, and Muhammad Irfan. “Entangled atomic ensemble and an yttrium-iron-garnet sphere in coupled microwave cavities”. In: *Physical Review A* 109.4 (Apr. 2024). ISSN: 2469-9934. DOI: [10.1103/PhysRevA.109.043708](https://doi.org/10.1103/PhysRevA.109.043708). URL: <http://dx.doi.org/10.1103/PhysRevA.109.043708>.
- [32] Jerad Inman et al. “Hybrid Magnonics for Short-Wavelength Spin Waves Facilitated by a Magnetic Heterostructure”. In: *Phys. Rev. Appl.* 17 (4 Apr. 2022), p. 044034. DOI: [10.1103/PhysRevApplied.17.044034](https://doi.org/10.1103/PhysRevApplied.17.044034). URL: <https://link.aps.org/doi/10.1103/PhysRevApplied.17.044034>.
- [33] Obed Alves Santos and Bart J. van Wees. “Magnon Confinement in an All-on-Chip YIG Cavity Resonator Using Hybrid YIG/Py Magnon Barriers”. In: *Nano Letters* 23.20 (2023). PMID: 37819876, pp. 9303–9309. DOI: [10.1021/acs.nanolett.3c02388](https://doi.org/10.1021/acs.nanolett.3c02388). URL: <https://doi.org/10.1021/acs.nanolett.3c02388>.
- [34] Yuzan Xiong et al. “Probing magnon–magnon coupling in exchange coupled $\text{Y}_3\text{Fe}_5\text{O}_{12}$ /Permalloy bilayers with magneto-optical effects”. In: *Scientific Reports* 10.1 (July 2020), p. 12548.

-
- ISSN: 2045-2322. DOI: [10.1038/s41598-020-69364-6](https://doi.org/10.1038/s41598-020-69364-6). URL: <https://doi.org/10.1038/s41598-020-69364-6>.
- [35] Kuldeep Kumar Shrivastava et al. “Unveiling photon-photon coupling induced transparency and absorption”. In: *Journal of Physics D Applied Physics* 57.46 (Aug. 2024), p. 465305. DOI: [10.1088/1361-6463/ad6613](https://doi.org/10.1088/1361-6463/ad6613). URL: <https://doi.org/10.1088/1361-6463/ad6613>.
- [36] Vahram Grigoryan and Jiang Xiao. “Dynamical spin-spin coupling of quantum dots”. In: *Europhysics Letters* 104.1 (Oct. 2013), p. 17008. DOI: [10.1209/0295-5075/104/17008](https://dx.doi.org/10.1209/0295-5075/104/17008). URL: <https://dx.doi.org/10.1209/0295-5075/104/17008>.
- [37] J. R. Schrieffer and P. A. Wolff. “Relation between the Anderson and Kondo Hamiltonians”. In: *Phys. Rev.* 149 (2 Sept. 1966), pp. 491–492. DOI: [10.1103/PhysRev.149.491](https://link.aps.org/doi/10.1103/PhysRev.149.491). URL: <https://link.aps.org/doi/10.1103/PhysRev.149.491>.
- [38] S. K. Samanta, R. Pradhan, and D. Syam. “Theoretical approach to verify the resonance frequency of a square split ring resonator”. In: *J. Opt. Soc. Am. B* 38.10 (Oct. 2021), pp. 2887–2897. DOI: [10.1364/JOSAB.428743](https://opg.optica.org/josab/abstract.cfm?URI=josab-38-10-2887). URL: <https://opg.optica.org/josab/abstract.cfm?URI=josab-38-10-2887>.
- [39] Harsimran Jit Kaur and Phalguni Goyal. “Comparison of two loop photonic crystal: ring-resonator with hexagonal and rectangular lattice structure”. In: *Journal of Optics* 50.2 (June 2021), pp. 223–236. ISSN: 0974-6900. DOI: [10.1007/s12596-020-00675-7](https://doi.org/10.1007/s12596-020-00675-7). URL: <https://doi.org/10.1007/s12596-020-00675-7>.
- [40] Sachin Verma et al. “Control of Photon-Magnon coupling in a Planar hybrid configuration”. In: *Journal of Superconductivity and Novel Magnetism* 37.5-7 (Mar. 2024), pp. 1163–1171. DOI: [10.1007/s10948-024-06721-w](https://doi.org/10.1007/s10948-024-06721-w). URL: <https://doi.org/10.1007/s10948-024-06721-w>.
- [41] Charles Kittel. “On the Theory of Ferromagnetic Resonance Absorption”. In: *Phys. Rev.* 73 (2 Jan. 1948), pp. 155–161. DOI: [10.1103/PhysRev.73.155](https://link.aps.org/doi/10.1103/PhysRev.73.155). URL: <https://link.aps.org/doi/10.1103/PhysRev.73.155>.
-

- [42] N. R. Bernier et al. “Level attraction in a microwave optomechanical circuit”. In: *Phys. Rev. A* 98 (2 Aug. 2018), p. 023841. DOI: [10.1103/PhysRevA.98.023841](https://doi.org/10.1103/PhysRevA.98.023841). URL: <https://link.aps.org/doi/10.1103/PhysRevA.98.023841>.
- [43] A. Metelmann and A. A. Clerk. “Quantum-Limited Amplification via Reservoir Engineering”. In: *Phys. Rev. Lett.* 112 (13 Apr. 2014), p. 133904. DOI: [10.1103/PhysRevLett.112.133904](https://doi.org/10.1103/PhysRevLett.112.133904). URL: <https://link.aps.org/doi/10.1103/PhysRevLett.112.133904>.
- [44] J. W. Rao et al. “Interactions between a magnon mode and a cavity photon mode mediated by traveling photons”. In: *Phys. Rev. B* 101 (6 Feb. 2020), p. 064404. DOI: [10.1103/PhysRevB.101.064404](https://doi.org/10.1103/PhysRevB.101.064404). URL: <https://link.aps.org/doi/10.1103/PhysRevB.101.064404>.
- [45] Kuldeep Kumar Shrivastava et al. *Emergence of coupling induced transparency by tuning purely dissipative couplings*. 2024. arXiv: [2409.12577](https://arxiv.org/abs/2409.12577) [quant-ph]. URL: <https://arxiv.org/abs/2409.12577>.
- [46] Dengke Zhang et al. “Observation of the exceptional point in cavity magnon-polaritons”. In: *Nature Communications* 8.1 (Nov. 2017), p. 1368. ISSN: 2041-1723. DOI: [10.1038/s41467-017-01634-w](https://doi.org/10.1038/s41467-017-01634-w). URL: <https://doi.org/10.1038/s41467-017-01634-w>.
- [47] Guo-Qiang Zhang and J. Q. You. “Higher-order exceptional point in a cavity magnonics system”. In: *Phys. Rev. B* 99 (5 Feb. 2019), p. 054404. DOI: [10.1103/PhysRevB.99.054404](https://doi.org/10.1103/PhysRevB.99.054404). URL: <https://link.aps.org/doi/10.1103/PhysRevB.99.054404>.
- [48] Patrick Kofod Mogensen and Asbjørn Nilsen Riseth. “Optim: A mathematical optimization package for Julia”. In: *Journal of Open Source Software* 3.24 (2018), p. 615. DOI: [10.21105/joss.00615](https://doi.org/10.21105/joss.00615).
- [49] Guido Menichetti, Matteo Calandra, and Marco Polini. “Electrical tuning of the magnetic properties of two-dimensional magnets: Cr₂Ge₂Te₆”. In: *Phys. Rev. B* 109 (20 May 2024), p. 205139. DOI: [10.1103/PhysRevB.109.205139](https://doi.org/10.1103/PhysRevB.109.205139). URL: <https://link.aps.org/doi/10.1103/PhysRevB.109.205139>.

-
- [50] Guido Menichetti, Matteo Calandra, and Marco Polini. “Electronic structure and magnetic properties of few-layer Cr₂Ge₂Te₆: the key role of nonlocal electron–electron interaction effects”. In: *2D Materials* 6.4 (Sept. 2019), p. 045042. DOI: [10.1088/2053-1583/ab2f06](https://doi.org/10.1088/2053-1583/ab2f06). URL: <https://dx.doi.org/10.1088/2053-1583/ab2f06>.
- [51] Sayandip Ghosh et al. “Plasmon-magnon interactions in two-dimensional honeycomb magnets”. In: *Phys. Rev. B* 107 (19 May 2023), p. 195302. DOI: [10.1103/PhysRevB.107.195302](https://link.aps.org/doi/10.1103/PhysRevB.107.195302). URL: <https://link.aps.org/doi/10.1103/PhysRevB.107.195302>.
- [52] T. J. Silva et al. “Inductive measurement of ultrafast magnetization dynamics in thin-film Permalloy”. In: *Journal of Applied Physics* 85.11 (June 1999), pp. 7849–7862. ISSN: 0021-8979. DOI: [10.1063/1.370596](https://doi.org/10.1063/1.370596). URL: <https://doi.org/10.1063/1.370596>.
- [53] Lihui Bai et al. “Spin Pumping in Electrodynamically Coupled Magnon-Photon Systems”. In: *Phys. Rev. Lett.* 114 (22 June 2015), p. 227201. DOI: [10.1103/PhysRevLett.114.227201](https://link.aps.org/doi/10.1103/PhysRevLett.114.227201). URL: <https://link.aps.org/doi/10.1103/PhysRevLett.114.227201>.
- [54] Yutaka Tabuchi et al. “Hybridizing Ferromagnetic Magnons and Microwave Photons in the Quantum Limit”. In: *Phys. Rev. Lett.* 113 (8 Aug. 2014), p. 083603. DOI: [10.1103/PhysRevLett.113.083603](https://link.aps.org/doi/10.1103/PhysRevLett.113.083603). URL: <https://link.aps.org/doi/10.1103/PhysRevLett.113.083603>.
- [55] Jinwei Rao et al. “Meterscale Strong Coupling between Magnons and Photons”. In: *Phys. Rev. Lett.* 131 (10 Sept. 2023), p. 106702. DOI: [10.1103/PhysRevLett.131.106702](https://link.aps.org/doi/10.1103/PhysRevLett.131.106702). URL: <https://link.aps.org/doi/10.1103/PhysRevLett.131.106702>.
- [56] R. D. McKenzie et al. “Theory of magnon polaritons in quantum Ising materials”. In: *Phys. Rev. A* 106 (4 Oct. 2022), p. 043716. DOI: [10.1103/PhysRevA.106.043716](https://link.aps.org/doi/10.1103/PhysRevA.106.043716). URL: <https://link.aps.org/doi/10.1103/PhysRevA.106.043716>.
-

- [57] I.A. Golovchanskiy et al. “Approaching Deep-Strong On-Chip Photon-To-Magnon Coupling”. In: *Physical Review Applied* 16.3 (Sept. 2021). ISSN: 2331-7019. DOI: [10.1103/PhysRevApplied.16.034029](https://doi.org/10.1103/PhysRevApplied.16.034029). URL: <http://dx.doi.org/10.1103/PhysRevApplied.16.034029>.
- [58] Yuzan Xiong et al. “Tunable Magnetically Induced Transparency Spectra in Magnon-Magnon Coupled $\text{Y}_3\text{Fe}_5\text{O}_{12}$ /Permalloy Bilayers”. In: *Phys. Rev. Appl.* 17 (4 Apr. 2022), p. 044010. DOI: [10.1103/PhysRevApplied.17.044010](https://doi.org/10.1103/PhysRevApplied.17.044010). URL: <https://link.aps.org/doi/10.1103/PhysRevApplied.17.044010>.
- [59] Kamran Ullah, M. Tahir Naseem, and Özgür E. Müstecaplıoğlu. “Tunable multiwindow magnomechanically induced transparency, Fano resonances, and slow-to-fast light conversion”. In: *Phys. Rev. A* 102 (3 Sept. 2020), p. 033721. DOI: [10.1103/PhysRevA.102.033721](https://doi.org/10.1103/PhysRevA.102.033721). URL: <https://link.aps.org/doi/10.1103/PhysRevA.102.033721>.
- [60] Samanvaya S. Gaur and Ernesto E. Marinero. “Emergent magnonic Materials: Challenges and opportunities”. In: *Materials* 16.18 (Sept. 2023), p. 6299. DOI: [10.3390/ma16186299](https://doi.org/10.3390/ma16186299). URL: <https://doi.org/10.3390/ma16186299>.
- [61] Sachin Verma et al. *Hybrid Photon-magnon Systems: Exploring the Purcell Effect*. 2025. arXiv: [2501.04574](https://arxiv.org/abs/2501.04574) [quant-ph]. URL: <https://arxiv.org/abs/2501.04574>.
- [62] Jeff Bezanson et al. “Julia: A Fresh Approach to Numerical Computing”. In: *SIAM Review* 59.1 (2017), pp. 65–98. DOI: [10.1137/141000671](https://doi.org/10.1137/141000671). eprint: <https://doi.org/10.1137/141000671>. URL: <https://doi.org/10.1137/141000671>.